



Project Title:- Smart Bus Scheduling System for Depot-Level Operations Using AI and Real-Time Data Analytics

Problem Statement:

Currently, bus schedule allocation at the depot is performed manually, which heavily depends on highly skilled and experienced personnel to select the right bus for the right schedule. The process is complex due to multiple operational constraints such as vehicle availability, battery charging status, washing completion, maintenance requirements, and route-specific needs.

Objective:

AI-based Smart Bus Scheduling System that can automatically analyze vehicle status, depot operations, and schedule requirements to allocate the most suitable buses, proactively manage operational constraints, and ensure on-time departures with improved fleet efficiency.

Schedule based on:

- Battery SOC & Expected range
- Schedule Distance
- Vehicle availability
- Charging status
- Driver availability

Proposed Solution:

An AI-powered Smart Bus Scheduling Platform that continuously analyzes real-time bus status against schedule requirements and automatically recommends the most suitable available bus for fleet operations.

"An AI-driven scheduling system that intelligently matches available buses with operational schedules in real time, enabling optimized fleet utilization and reliable depot operations."



Bus Depot Operations Process Map

Bus Charging

Bus Interior
Cleaning

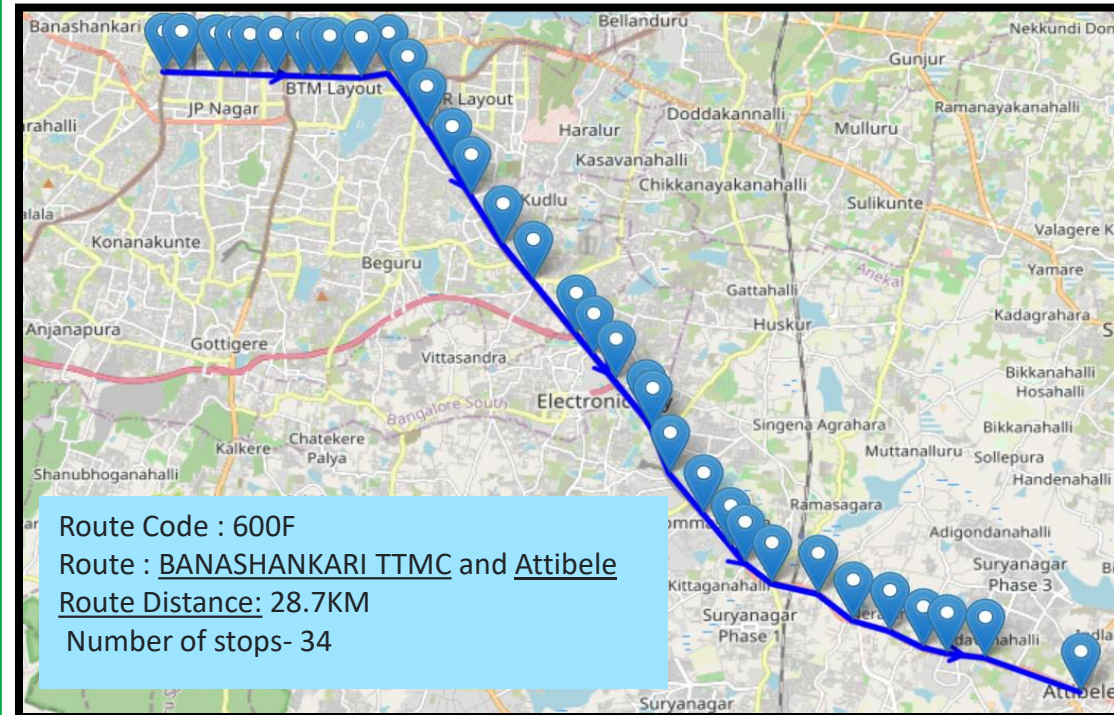
Bus Exterior
Cleaning

Schedule
Allocation



Inference: Project scope in the Schedule Allocation Process

A photograph showing a line of purple and white Switch Electric Buses parked on a street. The buses are modern, low-floor vehicles with large windows. The word "SWITCH" is visible on the side of the bus in the foreground. The background shows trees and a clear sky.



Route Code : 600F
Route : BANASHANKARI TTMC and Attibele
Route Distance: 28.7KM
Number of stops- 34





Schedule Allocation Guidelines



1. SOC(State Of Charge) 98-100%
2. Interior Cleaning
3. Exterior Cleaning
4. Bus- Route – Driver matching on right time

Bus Range Category	Meaning
A- High	>120 km
B- Medium	100-120
C- Low	<100

Route Category	Meaning
Simple Route	Easy to operate
Moderate Route	Normal effort
Complex Route	Requires additional planning

Time Slot	Category
05:00–07:00	Normal
07:00–10:00	Extreme Peak
10:00–16:00	Peak
16:00–20:00	Extreme Peak
20:00–23:00	Peak

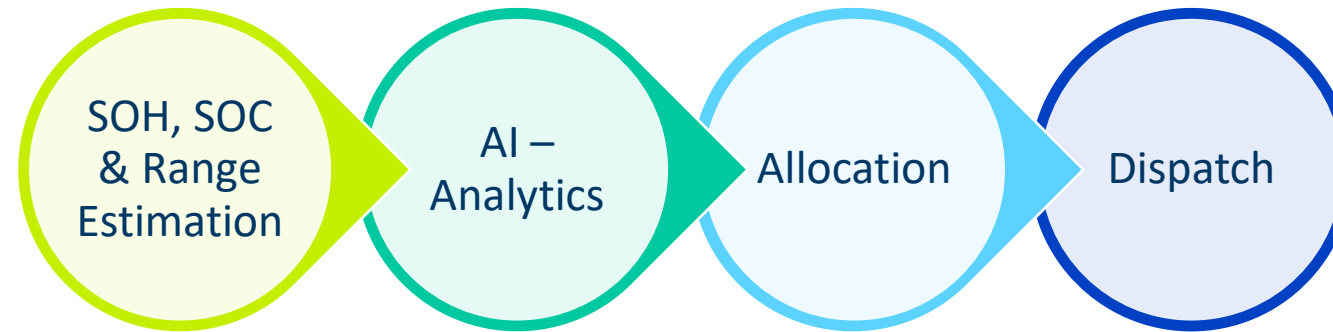
Route	Normal	Peak	Extreme Peak
Simple	A/B/C	A/B	A/B
Moderate	A/B	A	A
Complex	A	A	A

Inference: Bus Allocation based on SOH, SOC, Range & Availability





Proposed Solution



Inference: AI based Bus Allocation system to helps to choice right bus in right time without manual intervention



Project Title:- AI-Based Defect Capture & Prioritization System for 7 Depots

Problem Statement:

Defect capturing is currently managed through manual controls across depots, resulting in delayed reporting, inconsistent data, and limited visibility of recurring issues. Depot teams struggle to identify the highest contributing defects and prioritize corrective actions, leading to repeat failures, increased downtime, and reduced fleet availability.

Proposed solution:

Deploy an AI-powered defect analytics platform that captures defects from all depots, analyzes trends, automatically identifies the top defect contributors for each depot, and generates daily action plans through a centralized dashboard.

Objective:

- Automate defect capture across 7 depots
- Identify top contributing defects daily
- Prioritize corrective actions using AI
- Reduce manual effort and reporting errors
- Improve fleet reliability and availability
- Enable proactive maintenance planning

Defect management system in Depot



Current Situation	Action Plan	Benefits	Support Required	Remarks
- Defects are tracked manually, resulting in delayed identification, slow closure, repeat issues, and limited visibility of actions. - Resulted Penalty in All Depots	Implement a Defect Management System to log, track, assign, prioritize, and monitor defects.	- Faster defect identification and closure. - Reduction in recurring defects. - Improved product quality and customer satisfaction. - Better traceability and accountability.	- Management approval and project sponsorship. IT support for software development/implementation.	Start with a pilot in critical areas and expand across all depots

Exterior PDI check Sheet

Observation	Front LH	Front RH	Windshield Front	Head Lamp	Lamps	Body damages LH Side Door & lock	Glass damages Window Glass LH	Lamps	Indicator LH	Passenger Door Front	Passenger Door Rear	Rear LH	Rear RH	Rear Door	Windshield Rear	Tail Lamp	Body damages RH Side Door & lock	Glass damages Window Glass RH	Lamps	Indicator RH	Total Defect
#	Bus no	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front	Front
1	0403	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	05
2	0576	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	04
3	9982	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	03
4	4186	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
5	4380	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	04
6	0399	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	05
7	4160	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	02
8	1680	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	07
9	4178	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	03
10	0773	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
11	9989	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	04
12	3637	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
13	0395	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
14	9977	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
15	4186	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
16	0766	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	07
17	0399	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	07
18	0390	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	09
19	0393	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	09
20	4172	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
21	9976	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	06
22	0400	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	09
23	0391	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	05
24	0180	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	07
25	9973	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	01

Preview in Expo 88	
Preview	
DEPOT // OPS	
DEFECT DASH	
FLEET STATUS	
TOTAL DEFECTS LOGGED	
OPEN	CLOSED
0	1
HIGH SEV	MED SEV
1	0
BY DEPOT	
CENTRAL DEPOT	
TOTAL 1	
OPEN 0	CLOSED 1
NORTH DEPOT	
TOTAL 0	
OPEN 0	CLOSED 0
SOUTH DEPOT	
TOTAL 0	
+ LOG DEFECT	
TOP BUSES BY OPEN	
88 DASH	
DEFECTS	
LOG	
MANAGE	